KAIZEN MATHS

Lesson starter

Algebraic Simplification

Two Variables and Brackets

Name _____

S1

Simplify:

$4x + 3y + 4x + y$

S2

Simplify:

$-x - y - 5 + x - 8$

S3

Simplify:

$7(9a - 9) + 4(9a - 8)$

S4

Simplify:

$-x + 3y - 3x - 4y$

Show your working

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ALG-3B24-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Algebraic Simplification

Two Variables and Brackets

Name _____

E1

Simplify:

$-4x + 4y - 3 + 2x + 5$

E2

Simplify:

$3(4p + 1) + 2(9p + 3)$

E3

Simplify:

$-4x + 2y + 5x + y$

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ALG-3B24-L2 · E

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ALG-3B24-L2 · E

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ALG-3B24-L2 · E

Show your working

Teacher answers & working

Algebraic Simplification · Two Variables and Brackets

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S1

Simplify:

$4x + 3y + 4x + y$

Answer:

$8x + 4y$

Working:

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$4x + 3y + 4x + y = (4 + 4)x + (3 + 1)y$

$= 8x + 4y$

Step 2: Combined like terms for x and y separately.

Step 3: Write each simplified group once in the final expression.

S2

Simplify:

$-x - y - 5 + x - 8$

Answer:

$-y - 13$

Working:

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$-x - y - 5 + x - 8 = (-1 + 1)x + -y + (-5 + -8)$

$= -y - 13$

Step 2: Grouped x, y, and constants before combining.

Step 3: Write each simplified group once in the final expression.

S3

Simplify:

$7(9a - 9) + 4(9a - 8)$

Answer:

$99a - 95$

Working:

Step 1: Expand every bracket first by multiplying each term inside the bracket.

$7(9a - 9) + 4(9a - 8)$

$= 63a - 63 + 36a - 32$

$= (63 + 36)a + (-63 - 32)$

$= 99a - 95$

Step 2: The bracket multipliers are constants, so the powers of a do not change.

Step 3: Collect the like terms and write the simplified expression.

Teacher answers & working

Algebraic Simplification · Two Variables and Brackets

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S4

Simplify:

$-x + 3y - 3x - 4y$

Answer:

$-4x - y$

Working:

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$-x + 3y - 3x - 4y = (-1 + -3)x + (3 + -4)y$

$= -4x - y$

Step 2: Combined like terms for x and y separately.

Step 3: Write each simplified group once in the final expression.

E1

Simplify:

$-4x + 4y - 3 + 2x + 5$

Answer:

$-2x + 4y + 2$

Working:

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$-4x + 4y - 3 + 2x + 5 = (-4 + 2)x + 4y + (-3 + 5)$

$= -2x + 4y + 2$

Step 2: Grouped x, y, and constants before combining.

Step 3: Write each simplified group once in the final expression.

E2

Simplify:

$3(4p + 1) + 2(9p + 3)$

Answer:

$30p + 9$

Working:

Step 1: Expand every bracket first by multiplying each term inside the bracket.

$3(4p + 1) + 2(9p + 3)$

$= 12p + 3 + 18p + 6$

$= (12 + 18)p + (3 + 6)$

$= 30p + 9$

Step 2: The bracket multipliers are constants, so the powers of p do not change.

Step 3: Collect the like terms and write the simplified expression.

Teacher answers & working

Algebraic Simplification · Two Variables and Brackets

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E3

Simplify:

$-4x + 2y + 5x + y$

Answer:

$x + 3y$

Working:

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$-4x + 2y + 5x + y = (-4 + 5)x + (2 + 1)y$

$= x + 3y$

Step 2: Combined like terms for x and y separately.

Step 3: Write each simplified group once in the final expression.